Math 133 Written Homework 1

Due Thursday, September 10, 2026, at 11:59 p.m.

Submit one legible PDF on Gradescope, clearly identify each problem, and assign the correct page(s) to every problem. Show enough reasoning that another student in the course could follow your work, and include units whenever they are meaningful. You may discuss problems with classmates, but your submitted work must be your own. Late work will not be accepted.

0. **Assignment record.**

   (a) List any collaborators and the problem(s) you discussed with each person. If none, write "None."

   (b) How much time did you spend on this assignment?

   (c) Disclose any use of generative AI: name the tool and describe how you used it. If none, write "None."

1. **A product launch, viewed through functions.**

   An online retailer tracked the price and quantity sold of a new product during its first six weeks. Let $P(w)$ be the price in week w, and let $Q(w)$ be the number of units sold in week w.

w (week)	1	2	3	4	5	6
$P(w)$ (dollars per unit)	18	20	22	20	19	21
$Q(w)$ (units)	420	380	330	395	410	360

   (a) State the domain and range of both P and Q. Use set notation and include units.

   (b) Explain why P and Q are functions of w. Is Q a function of P for these data? Justify your answer using specific table entries.

   (c) Define weekly revenue by $R(w) = P(w)Q(w)$. Complete a row of values for R, then state its domain and range. In which week was revenue largest?

   (d) A manager claims, "Whenever we charged a lower price, our weekly revenue was higher." Does the table support this claim? Give a specific comparison that settles the question.

   (e) Is R a function of P for these data? Explain. Then describe one reason that a six-week table cannot, by itself, establish that changing the price *caused* the change in sales.

2. **The average cost of a market-day candle.**

   A candle studio pays \$180 to rent a market booth and spends \$3.60 in materials for each candle it makes. The studio can make at most 120 candles in one day. If q candles are produced, define

   $$C(q) = 180 + 3.60q \qquad \text{and} \qquad A(q) = \frac{C(q)}{q}.$$

   (a) Identify the input and output of each function, including units. What is the realistic, contextual domain of C?

   (b) Explain why $C(0)$ is defined but $A(0)$ is not. State both the natural algebraic domain of A and its realistic, contextual domain.

   (c) Calculate $A(12)$ and $A(60)$. Interpret each answer in a complete sentence.

   (d) Write the exact contextual range of A in set-builder notation. Find its minimum and maximum values.

   (e) What is the smallest number of candles the studio can make so that its average cost is at most \$6 per candle?

3. **A two-tier delivery model.**

   A local courier models the pre-surcharge price of a delivery by

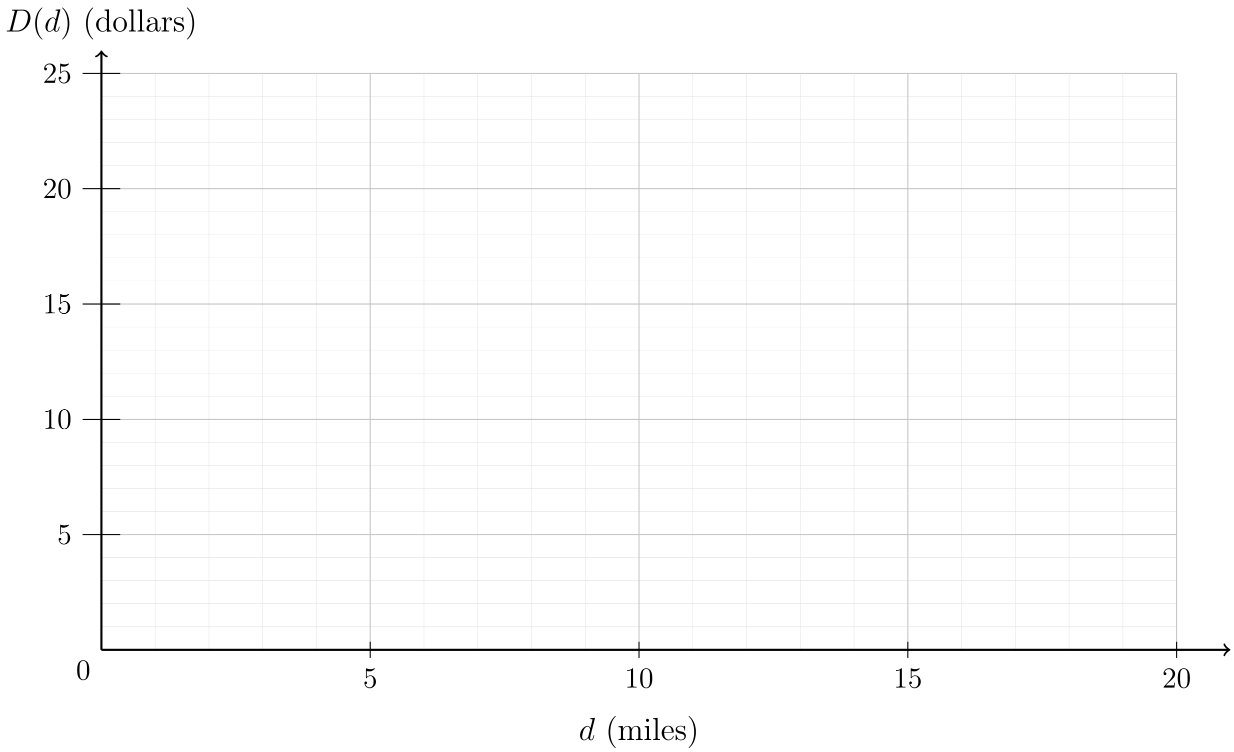
   $$D(d) = \begin{cases} 7.50 + 1.20d, & 0 \le d \le 5, \\ 10.50 + 0.60d, & 5 < d \le 20, \end{cases}$$

   where d is the delivery distance in miles and $D(d)$ is measured in dollars. The delivery platform then applies a 7.75% surcharge, represented by $T(c) = 1.0775c$.

   (a) State the contextual domain and range of D, with units.

   (b) Calculate and interpret $D(0)$, $D(5)$, and $D(12)$.

(c) Draw a carefully labeled graph of D. Mark the endpoints and the point where the pricing rule changes. Is D increasing, decreasing, or neither on its domain?



(d) Verify algebraically that there is no price jump at five miles. Explain what *does* change there.

(e) Let $F = T \circ D$. Find a piecewise formula for $F(d)$ and calculate the final charge for a 12-mile delivery. Round only the final monetary answer to the nearest cent.

4. **Spreadsheet investigation: 30 daily S&P 500 values.**

The S&P 500 is an index that summarizes the market value of a collection of large U.S. companies. Each entry in the `SP500` column is a scaled *index level* measured in index points, not dollars or shares. For example, a value of 7,500 means that the index level is 7,500 points; it does not mean $7,500. A one-point change in the index has no fixed dollar value. The Federal Reserve Bank of St. Louis provides the data used in this problem.

Get the data. Download the prepared CSV file and open it in Excel or Google Sheets. The file already contains exactly 30 observations, arranged from July 21 through August 31, 2026. It has only two columns: `observation_date` and `SP500`. Do not filter, delete, or rearrange the data.

(a) **Open and format the data.** Rename the sheet `S&P 500`. Verify that the headings are in row 1 and the observations are in rows 2–31. Format column A as dates and column B as numbers with two decimal places.

(b) **Create calculated columns.** Add these headings:

Column C	Column D	Column E
`Daily Point Change`	`Daily Percent Change`	`Value Index`

Because July 21 has no preceding day in the file, enter `N/A` in cells C2 and D2. Then enter

`C3 = B3-B2`, `D3 = (B3-B2)/B2`, `E2 = 100*B2/$B$2`.

Use Quick Fill to copy C3 and D3 through row 31, and E2 through row 31. Format column D as a percentage with two decimal places and column E as a number with two decimal places. Verify that E2 is 100. Explain in one sentence what an index value of 102 means.

(c) **Summarize the period.** Use spreadsheet formulas to find:

- the average S&P 500 value: `=AVERAGE(B2:B31)`;
- the smallest value: `=MIN(B2:B31)`;
- the largest value: `=MAX(B2:B31)`;
- the total point change: `=SUM(C3:C31)`.

Record the date of the smallest value and the date of the largest value. Check your total point change by calculating `=B31-B2`; the two answers should agree.

(d) **Graph the data.** Make two scatterplots with connected markers:

- `observation_date` versus `SP500`;
- `observation_date` versus `Daily Percent Change`.

Give each graph a descriptive title and labeled axes. Make the dates readable and use a sensible viewing window. Beneath each graph, write one sentence describing something visible in that graph.

Include with Problem 4 in your Gradescope PDF:

- a screenshot of the completed spreadsheet, including the summary calculations;
- the formulas used in the first calculation row;
- screenshots of both graphs and the two sentences describing them.